



10 million reasons

As malicious Android apps hitting 10 Million, Kaspersky has some advice for Android users to protect against malignant cyber-scammers <http://dgit.in/maland>

Alt+Tabbing troubles

If you've always wondered why using Alt+Tab combination while you're playing a game in fullscreen mode, freezes or slows down your PC. Read this: <http://dgit.in/AltTabbing>



Graphene Aerogel on a plant spine

Apart from comfortably sitting atop flowers, Graphene Aerogel has been stated as having very high oil absorption capabilities, and so could be a natural remedy for an oil spill. Additionally, the researchers have claimed that it is recyclable too.

The Heaviest

For years and year researchers have tried to prove the existence of element 118, the heaviest element known to man temporarily called ununoctium, but since the stability of this element is so bizarre it has yet not formally taken its place among the other elements. After some researchers proved that it exists in 1999, a few years later they had to conclude that their proof was based on some miscalculations and there were talks that a researcher had intentionally written bogus results. Element 118 did not exist. In 2006 though, researchers from US based Lawrence Livermore collaborated with Russian Joint Institute for Nuclear Research managed to create element 118 again.

The element was created by the bombardment of calcium ions into a target of californium, an artificial element with 98 protons, for three months continuously. To get the desired result, it was required that the elements being bombarded collide head on with a specific energy, a very rare phenomenon, failing to do which the

elements would just coolly pass through each other.

After all of this, only a mere three collisions of ions caused the infamous element 118 to be created.

What makes the element more mysterious is that it hasn't really been seen as such. After the required collisions, element 118 was created but it lasted for less than a millisecond. The only reason the researchers were able to find out that they had created the element was because they were able to study what the element broke down into and work their way upwards.

The researchers have said that the new element could help in the understanding of nuclear stability and the fission process. Another reason is what has been called 'the island of stability'. Scientists have been creating heavier and heavier elements since 1940, and they realized that as the elements become heavier they also became much more unstable. But weirdly enough some elements with a specific higher weight seem to be more stable than the others. Building element 118, or heavier and heavier elements, is one of the key ways to find out more about this island of stability. With a longer shelf life, heavier elements can be used in more and more applications.

And if you are wondering why we are talking about an element which was discovered in 2006, know that element

118 is just on its way to its new home on the periodic table, even though it was discovered ten years ago. The process for certifying a new element is a long one.

A year ago, researchers in Germany had attempted to creating element 119. The bombardment process is over- and right now they are sifting through their data to find any trace of element 119- and in the process make a high school student's life that much... heavier.

The Strongest

So the the award for the strongest material known to man was for a long time given to Graphene, a truly remarkable material with some very interesting characteristics over which people are still falling over each other wondering how to put it to commercial use. And then enters Carbyne, with its own set of characteristics which makes Graphene feel like the once-famous kid-on-the-block. Carbyne's initial journey began way back in 1960, although there was not much known about it for until a few months ago when Rice University came out with its research stating that Carbyne truly was the strongest material known to man currently. Carbyne is a chain of carbon atoms held together by either double or alternating single and triple atomic bonds making it a one-dimensional material having only length, unlike Graphene which is more of a two-dimensional material having length and width. It is this characteristic of Carbyne which scientists believe gives it its peculiar mechanical and electrical properties.

Carbyne's tensile strength has been said to be double that of graphene; its tensile stiffness is said to be twice that of graphene and carbon nanotube and almost three times as much as a diamond's; and a 90 degree rotation of the material transforms it into a magnetic semi-conductor.

The only problem now with regard to Carbyne is in its mass production. Graphene is still not actively used in any major applications, although its synthesis has for now become much easier. Scientists hope that the same may happen to Carbyne- which currently can do so much but will have to wait for some time for the rest of the world to catch up with it. 